

kernel/syscall.c

```

124 [SYS_write]    sys_write,
125 [SYS_mknod]   sys_mknod,
126 [SYS_unlink]  sys_unlink,
127 [SYS_link]    sys_link,
128 [SYS_mkdir]   sys_mkdir,
129 [SYS_close]   sys_close,
130 [SYS_interpose] sys_interpose,
131 };
132
133 void
134 syscall(void)
135 {
136     int num;
137     struct proc *p = myproc();
138
139     num = p->trapframe->a7;
140     if (num > 0 && num < NELEM(syscalls) && syscalls[num]) {
141         if ((p->deny_mask >> num) & 1) {
142             // denied: return -1 to user
143             p->trapframe->a0 = -1;
144             return;
145         }
146         p->trapframe->a0 = syscalls[num]();
147     } else {
148         printf("%d %s: unknown sys call %d\n", p->pid, p->name, num);
149         p->trapframe->a0 = -1;
150     }

```

B+

>

remote Thread 1.3 (src) In: syscall

L137 PC: 0x80001d10

#1 0x0000000080001ac8 in usertrap () at kernel/trap.c:68

#2 0x0000003fffffff09c in ?? ()

(gdb) n

(gdb) p/x *p

value has been optimized out

(gdb) p/x p->trapframe->a7

value has been optimized out

(gdb) p/x \$sstatus

\$1 = 0x200000022

(gdb) p/x myproc()->trapframe->a7

\$2 = 0xf

(gdb) █

```

usys.o user/printf.o user/umalloc.o
riscv64-elf-objdump -S user/_secret > user/secret.asm
riscv64-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind us
er/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_se
cret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1063 blocks have been allocated
balloc: write bitmap block at sector 46
*** Now run 'gdb' in another window.
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global
virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-devic
e,drive=x0,bus=virtio-mmio-bus.0 -S -gdb tcp::25504
qemu-system-riscv64: QEMU: Terminated via GDBstub
(base) hamza@krm14 xv6-labs-2025 (Lab2) % make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind us
er/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_se
cret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1063 blocks have been allocated
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qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global
virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-devic
e,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
scause=0xd sepc=0x80001d14 stval=0x0
panic: kerneltrap

```



```

Last login: Sat Oct  4 16:35:00 on ttys001
(base) hamza@krm14 ~ % cd '/Users/hamza/Documents/C_Coding/xv6-labs-2025 (Lab2)'
(base) hamza@krm14 xv6-labs-2025 (Lab2) % riscv64-elf-gdb kernel/kernel
GNU gdb (GDB) 16.3
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "--host=aarch64-apple-darwin25.0.0 --target=riscv64-elf".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from kernel/kernel...
The target architecture is set to "riscv:rv64".
.gdbinit:3: Error in sourced command file:
could not connect: Operation timed out.
(gdb) target remote :26000
Remote debugging using :26000
0x0000000000000100 in ?? ()
(gdb) b syscall
Breakpoint 1 at 0x8001d04: file kernel/syscall.c, line 135.
(gdb) c
Continuing.

Thread 1 hit Breakpoint 1, syscall () at kernel/syscall.c:135
135      {
(gdb) b *0xYOUR_SEPC
Invalid number "0xYOUR".
(gdb) b *0xf
Breakpoint 2 at 0xf
(gdb) layout asm

```

```
B+> 0x80001d04 <syscall>      addi      sp, sp, -32
0x80001d06 <syscall+2>      sd          ra, 24(sp)
0x80001d08 <syscall+4>      sd          s0, 16(sp)
0x80001d0a <syscall+6>      sd          s1, 8(sp)
0x80001d0c <syscall+8>      addi      s0, sp, 32
0x80001d0e <syscall+10>     jal      0x80000d8e <myproc>
0x80001d12 <syscall+14>     mv          s1, a0
0x80001d14 <syscall+16>     lw          a3, 0(zero) # 0x0
0x80001d18 <syscall+20>     addiw     a4, a3, -1
0x80001d1c <syscall+24>     li          a5, 21
0x80001d1e <syscall+26>     bltu     a5, a4, 0x80001d4e <syscall+74>
0x80001d22 <syscall+30>     slli      a4, a3, 0x3
0x80001d26 <syscall+34>     auipc     a5, 0x6
0x80001d2a <syscall+38>     addi      a5, a5, -1462
0x80001d2e <syscall+42>     add       a5, a5, a4
0x80001d30 <syscall+44>     ld        a4, 0(a5)
0x80001d32 <syscall+46>     beqz     a4, 0x80001d4e <syscall+74>
0x80001d34 <syscall+48>     ld        a5, 208(a0)
0x80001d36 <syscall+50>     srl      a5, a5, a3
0x80001d3a <syscall+54>     andi     a5, a5, 1
0x80001d3c <syscall+56>     bnez     a5, 0x80001d46 <syscall+66>
0x80001d3e <syscall+58>     ld        s1, 88(a0)
0x80001d40 <syscall+60>     unimp
0x80001d42 <syscall+62>     unimp
0x80001d44 <syscall+64>     unimp
0x80001d46 <syscall+66>     unimp
0x80001d48 <syscall+68>     unimp
0x80001d4a <syscall+70>     unimp
0x80001d4c <syscall+72>     unimp
0x80001d4e <syscall+74>     unimp
0x80001d50 <syscall+76>     unimp
0x80001d52 <syscall+78>     unimp
0x80001d54 <syscall+80>     unimp
0x80001d56 <syscall+82>     unimp
```

remote Thread 1.1 (asm) In: syscall

L135 PC: 0x80001d04

(gdb) c

Continuing.

(gdb) █


```
usys.o user/printf.o user/umalloc.o
riscv64-elf-objdump -S user/_secret > user/secret.asm
riscv64-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind us
er/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_se
cret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1063 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global
virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-devic
e,drive=x0,bus=virtio-mmio-bus.0
```

xv6 kernel is booting

hart 1 starting

hart 2 starting

init: starting sh

\$ sandbox 32768 - cat README

cat: cannot open README

\$ sandbox 32768 README grep xv6 README

xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix Version 6 (v6). xv6 loosely follows the structure and style of v6, xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer (kaashoek,rtm@mit.edu)). The main purpose of xv6 is as a teaching

\$ sandbox 32768 README grep xv6 x

grep: cannot open x

\$ sandbox 128 - ls

sandbox: exec ls failed

\$ sandbox 32896 - grep xv6 README

sandbox: exec grep failed

\$ sandbox 32768 - cat README

cat: cannot open README

\$

```
-fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/secret.o user/secret.c
riscv64-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_secret user/secret.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-elf-objdump -S user/_secret > user/secret.asm
riscv64-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1106 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
```

xv6 kernel is booting

hart 2 starting

hart 1 starting

init: starting sh

\$ secret abcDE12345

\$ attack

abcDE12345

\$ secret yello765

\$ attack

yello765

\$ attack

abcDE12345

\$ attack

yello765

\$


```
no-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-bu
iltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprin
tf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kall
oc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kern
el/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o
kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/
virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (4.7s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (1.0s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.2s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.8s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.1s)
== Test attack ==
$ make qemu-gdb
attack: OK (10.0s)
== Test time ==
time: OK
Score: 45/45
(base) hamza@krm14 xv6-labs-2025 (Lab2) %
```